Likelihood ratio tests

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Neyman-Pearson test

Let $\mathbf{X} = (X_1, ..., X_n)$ be a sample from some distribution with probability function f and parameter θ. To test

$$H_0 : \theta = \theta_0 \qquad H_A : \theta = \theta_1,$$

the **Neyman-Pearson** test rejects H_0 when

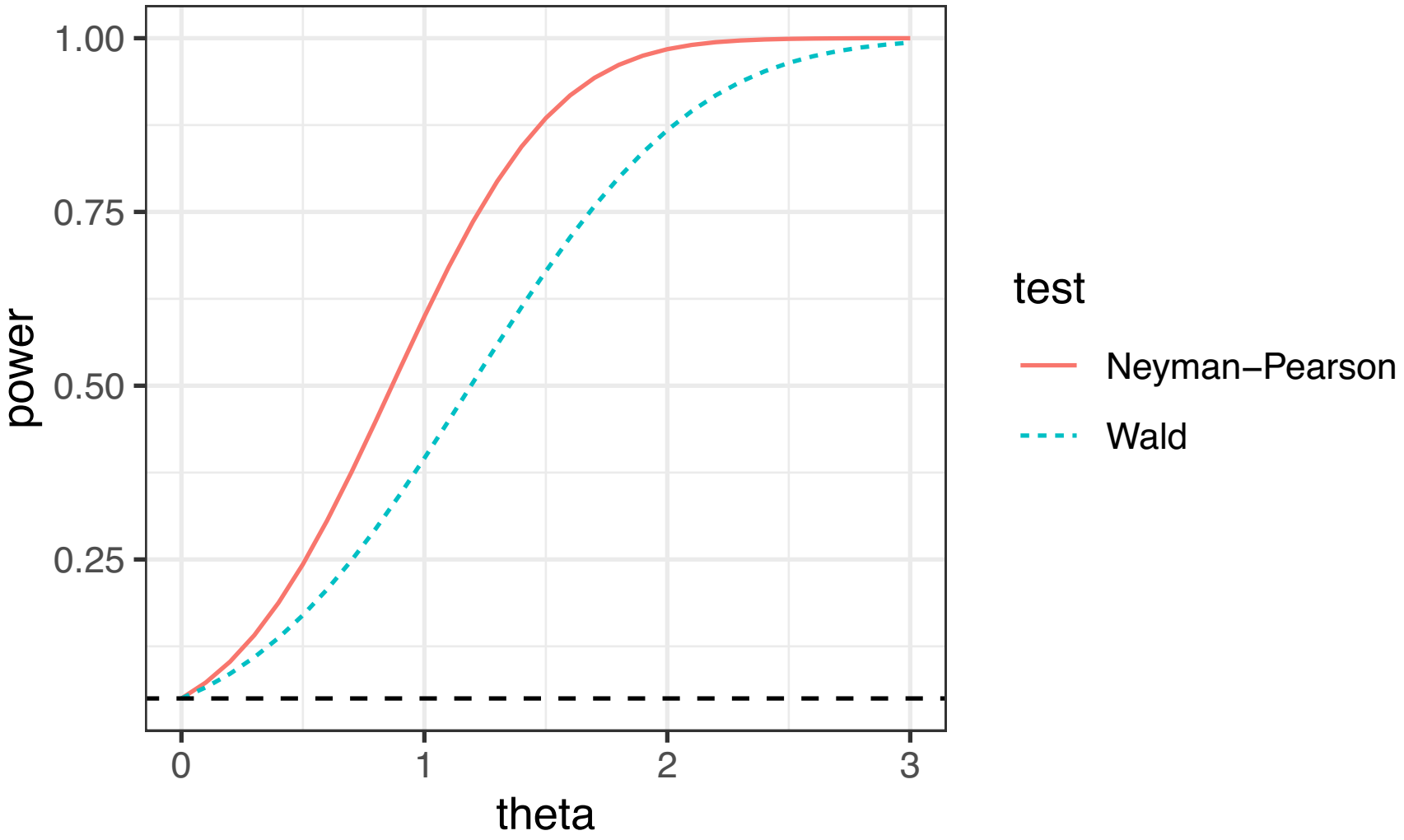
$$\frac{L(\theta_1|\mathbf{X})}{L(\theta_0|\mathbf{X})} = \frac{f(X_1, ..., X_n|\theta_1)}{f(X_1, ..., X_n|\theta_0)} > k$$

where k is chosen so that $\beta(\theta_0) = \alpha$.

Last time: Example

$$X_1, ..., X_{20} \overset{indep.}{\sim} N(\theta, \sigma_i^2) \text{ where } \sigma_i^2 = i$$

$$H_0 : \theta = 0 \qquad H_A : \theta = \theta_1 \quad \text{where } \theta_1 > 0$$



Most powerful test

Big idea: can't do better than a likelihood ratio test for two simple hypotheses!

What does it mean for one test to be "better" than another?

$P_\theta(\text{reject}) \le \alpha \quad \forall \theta \in \Theta_0$

Definition (UMP test): Consider testing $H_0 : \theta \in \Theta_0$ vs. $H_A : \theta \in \Theta_1$. Let $\mathcal{C}_\alpha$ be the set of all level α tests for these hypotheses. A test in $\mathcal{C}_\alpha$ is the **uniformly most powerful** (UMP) level α test if:

it has power $\beta(\theta) \ge \beta^*(\theta) \quad \forall \theta \in \Theta_1$

where β^* is the power for any other test in $\mathcal{C}_\alpha$

Neyman-Pearson setting: $H_0 : \theta = \theta_0 \quad H_A : \theta = \theta_1$

UMP level α test: $\beta(\theta_0) \le \alpha$

$\beta(\theta_1) \ge \beta^*(\theta_1)$

Neyman-Pearson lemma

Let $\mathbf{X} = (X_1, ..., X_n)$ be a sample from some distribution with probability function $f(x|\theta)$. Consider testing $H_0 : \theta = \theta_0$ vs. $H_A : \theta = \theta_1$.

Suppose we reject H_0 when

$$\frac{L(\theta_1|X)}{L(\theta_0|X)} > K$$

This is the UMP of the hypotheses of size $\alpha = \beta(\theta_0)$

(see C&B for proof)

Last time: Example

Let $X_1, ..., X_n \overset{iid}{\sim} Exponential(\theta)$, with pdf $f(x|\theta) = \theta e^{-\theta x}$. We want to test

$$H_0 : \theta = \theta_0 \qquad H_A : \theta = \theta_1,$$

where $\theta_1 < \theta_0$.

$$\frac{L(\theta_1|X)}{L(\theta_0|X)} > K \iff \sum_{i=1}^{n} X_i > c \quad \text{for some } c$$

$$c = F^{-1}_{n,\theta_0}(1-\alpha)$$

F_{n,θ_0} is cdf of Gamma(n, θ_0)

$$\Rightarrow \beta(\theta_0) = \alpha$$

$\Rightarrow$ The UMP level α test of these hypotheses rejects when $\sum_{i=1}^{n} X_i > F^{-1}_{n,\theta_0}(1-\alpha)$

<u>Note</u>: This test depends on θ_0, but not on θ_1!

Just need to know that $\theta_1 < \theta_0$

So for <u>any</u> $\theta_1 < \theta_0$, the UMP test reject when $\sum_i X_i > F^{-1}_{n,\theta_0}(1-\alpha)$

UMP tests of composite hypotheses

Consider testing $H_0: \theta = \theta_0$ vs. $H_A: \theta \in \Theta_1$

If a test is the UMP level α test of

$$H_0: \theta = \theta_0 \quad \text{vs.} \quad H_A: \theta = \theta_1$$

for <u>all</u> $\theta_1 \in \Theta_1$, and the rejection region of the test <u>doesn't</u> depend on θ_1, then it is also UMP level α test of

$$H_0: \theta = \theta_0 \quad \text{vs.} \quad H_A: \theta \in \Theta_1$$

<u>Ex</u>: $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exponential}(\theta)$

The UMP level α test of

$$H_0: \theta = \theta_0 \quad \text{vs.} \quad H_A: \theta < \theta_0$$

rejects when $\sum_i X_i > F^{-1}_{n,\theta_0}(1-\alpha)$

F_{n,θ_0} = cdf of Gamma(n, θ_0)

UMP tests of composite hypotheses

Definition (MLR): Let $\mathcal{P} = \{f(x|\theta) : \theta \in \Theta \subseteq \mathbb{R}\}$ be a family of pdfs or pmfs. We say that $\mathcal{P}$ has **monotone likelihood ratios** (MLR) in the statistic $T(X)$ if:

$$\frac{L(\theta_2|x)}{L(\theta_1|x)} = \frac{f(x|\theta_2)}{f(x|\theta_1)}$$ is a non-decreasing function of $T(x)$ for all θ_1, θ_2 w/ $\theta_2 > \theta_1$

Ex: $X_1, \dots, X_n \overset{iid}{\sim} N(\theta, \sigma^2)$

$$L(\theta|x) = (2\pi\sigma^2)^{-\frac{n}{2}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\theta)^2\right\}$$

$$\frac{L(\theta_2|x)}{L(\theta_1|x)} = \exp\left\{\frac{n\bar{x}(\theta_2-\theta_1)}{\sigma^2} - \frac{n}{2\sigma^2}(\theta_2^2-\theta_1^2)\right\}$$

increasing in $\bar{x}$ (or equiv. in $\sum_{i=1}^{n} x_i$) if $\theta_2 > \theta_1$

$\Rightarrow$ MLR in $\bar{x}$ (or equiv. in $\sum_{i=1}^{n} x_i$)

UMP tests of composite hypotheses

Definition (MLR): Let $\mathcal{P} = \{f(x|\theta) : \theta \in \Theta \subseteq \mathbb{R}\}$ be a family of pdfs or pmfs. We say that $\mathcal{P}$ has **monotone likelihood ratios** (MLR) in the statistic $T(X)$ if $\dfrac{f(x|\theta_2)}{f(x|\theta_1)}$ is a non-decreasing function of $T(x)$, for any pair $\theta_2 > \theta_1$.

Theorem: Let $\mathcal{P}$ be a family of distributions, and suppose we want to test $H_0 : \theta \leq \theta_0$ vs. $H_A : \theta > \theta_0$ for some $\theta_0 \in \mathbb{R}$.

If $\mathcal{P}$ has MLR in $T(X)$, then the test that rejects H_0 when $T(X) > t_0$ is UMP at level $\alpha = P_{\theta_0}(T(X) > t_0)$

<u>Intuition</u>: ① For any $\theta_1 > \theta_0$, rejection depends only on θ_0

② By N-P lemma, test is UMP for each $H_0: \theta = \theta_0$ vs. $H_A: \theta = \theta_1$

③ $\beta(\theta) \leq \beta(\theta_0)$ for $\theta \leq \theta_0$

Example

Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\theta, \sigma^2)$, and consider testing $H_0 : \theta \leq \theta_0$ vs. $H_A : \theta > \theta_0$.

Normal has MLR in $\bar{X}$

So: UMP test of $H_0: \theta \leq \theta_0$ vs. $H_A: \theta > \theta_0$

rejects when $\bar{X} > t_0$

$$\bar{X} \sim N\left(\theta, \frac{\sigma^2}{n}\right) \Rightarrow \frac{\bar{X} - \theta}{\sigma/\sqrt{n}} \sim N(0,1)$$

$$\frac{\bar{X} - \theta}{s/\sqrt{n}} \approx N(0,1)$$

$$P_{\theta_0}(\bar{X} > t_0) = P_{\theta_0}\left(\frac{\bar{X} - \theta_0}{s/\sqrt{n}} > \frac{t_0 - \theta_0}{s/\sqrt{n}}\right)$$

$$\approx 1 - \Phi\left(\frac{t_0 - \theta_0}{s/\sqrt{n}}\right) \overset{\text{set}}{=} \alpha$$

$$\Rightarrow t_0 = \Phi^{-1}(1-\alpha)\frac{s}{\sqrt{n}} + \theta_0$$

UMP test rejects when $\bar{X} > \Phi^{-1}(1-\alpha)\frac{s}{\sqrt{n}} + \theta_0$

Example

Solution ① $L(\lambda|X) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{e^{-n\lambda} \lambda^{\sum_i x_i}}{\prod_i x_i!}$

$$\Rightarrow \frac{L(\lambda_2|X)}{L(\lambda_1|X)} = e^{n(\lambda_1-\lambda_2)} \left(\frac{\lambda_2}{\lambda_1}\right)^{\sum_{i=1}^{n} x_i}$$

For $\lambda_2 > \lambda_1$, this is increasing in $\sum_{i=1}^{n} x_i$

so MLR in $\sum_{i=1}^{n} x_i$

Let $X_1, ..., X_n \overset{iid}{\sim} Poisson(\lambda)$, and let $f(x|\lambda)$ be the pmf.

1. Show that $\mathcal{P} = \{f(x|\lambda) : \lambda > 0\}$ has MLR in $\sum_{i=1}^{n} X_i$.
2. Find the UMP test of $H_0 : \lambda \leq \lambda_0$ vs. $H_A : \lambda > \lambda_0$.

② MLR in $T(X) = \sum_{i=1}^{n} X_i$, so reject H_0 when $\sum_{i=1}^{n} X_i > t_0$

Recall that since $X_1, ..., X_n \overset{iid}{\sim} Poisson(\lambda)$, then

$\sum_{i=1}^{n} X_i \sim Poisson(n\lambda)$

Choose t_0 such that $P_{\lambda_0}\left(\sum_{i=1}^{n} X_i > t_0\right) \leq \alpha$

Let F_{n,λ_0} be cdf of $Poisson(n\lambda_0)$

$\Rightarrow 1 - F_{n,\lambda_0}(t_0) \leq \alpha \Rightarrow t_0 = F^{-1}_{n,\lambda_0}(1-\alpha)$

So: UMP test rejects when $\sum_{i=1}^{n} x_i > F^{-1}_{n,\lambda_0}(1-\alpha)$